

coverage_ledger

BobaLink Browser Extension — Feature Coverage Ledger

Revision: 6 **Last modified:** 2026-06-13T12:48:00Z **Scope:** BobaLink (extension/) — feature × test-type coverage ledger with a per-feature autonomous-verification class (§11.4.25 full-automation-coverage, §11.4.52 autonomous-validation). **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (Phase 8, T8.5); live status docs/browser_extension/Status.md (Rev 4).

§11.4.6 (no-guessing) / §11.4.5 (captured evidence): every ✓ cell cites a REAL test file backing it. A cell with no real test file is **PENDING** — never silently marked covered. The e2e row is honestly OPERATOR_ATTENDED_ONLY in this headless sandbox: the MV3-load test is real but SKIPS (no extension service worker registers — §11.4.3 topology dispatch), so it is not autonomous here.

Ground-truth baseline (this session)

- Full suite: **814 passed / 68 spec files** (npx vitest run via extension/ci-ext.sh → CI-EXT: PASS, Session 12 wave-13 / 2026-06-13; +5 popup partial-Send-All gap + 2 prior-session files brought into git: options-save-flow (4) + offline-queue-persistence (6)) under extension/tests/{unit,perf,stress,chaos,integration,security,ally,i18n,e2e} + src/**/*. (+240 over the 559 rc baseline: +73 wave-10, +129 wave-11, +38 wave-12.)
- **Session-12 wave-12 coverage** (no product defects found in any): AES-256-GCM crypto tamper/auth/fixed-IV (tests/security/crypto-tamper.test.ts), LinkScanner scheme-allowlist/visibility/dedup (tests/unit/link-scanner-coverage.test.ts), content-script

highlight-manager DOM no-leak/idempotency/href-preservation (tests/unit/highlight-manager.test.ts).

- **Session-12 coverage breadth added** (closing prior PENDING ledger gaps; all anti-bluff, no absolute wall-clock thresholds):
 - **stress/chaos breadth** beyond parsers+queue — tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts (orchestrator junk-flood, real client FIFO-flood, rate-limiter engagement, tab-group at scale, chaos fault-injection).
 - **security — detection/infoshash paths** — tests/security/infoshash-detection-hostile.test.ts (hex/base32 boundary+case, xt= confusion, .torrent URL allowlist, SHA-1 infoshash correctness, hostile-repetition dedup).
 - **accessibility — focus + contrast (deeper WCAG)** — tests/ally/focus-and-contrast.ally.test.ts (WCAG 2.4.3/2.4.7/2.1.2 focus management + 1.4.3 computed contrast from the real CSS).
 - **DoS scaling (machine-independent)** — tests/perf/orchestrator-scaling.perf.test.ts (metamorphic sub-quadratic ratio + oracle self-validation).
- **Session-12 wave-11 coverage breadth added** (all anti-bluff, no absolute wall-clock thresholds):
 - **dynamic-content re-scan** — tests/unit/orchestrator-dynamic-content.test.ts (MutationObserver insert / href-mutation / relevance-filter / debounce-coalesce / stop-disconnect).
 - **i18n locale safety (8 locales)** — tests/i18n/locale-safety.test.ts (placeholder/key parity, XSS-inert values, JSON integrity).
 - **client resilience** — tests/chaos/boba-client-resilience.chaos.test.ts (retry / backoff / timeout / 4xx-vs-5xx classification under fault injection).
 - **bencode/.torrent parser robustness** — tests/security/bencode-torrentfile-hostile.test.ts (truncation, hostile lengths, integer edge forms, deep nesting, binary-safe infoshash).
 - **bare-text magnet detection** — tests/unit/text-scanner-coverage.test.ts (cross-scanner id-equality, false-positive boundary, non-content-node skipping).
 - **options-page a11y** — tests/ally/options-contrast-motion.ally.test.ts (contrast + focus + reduced-motion; found+fixed 5 real WCAG defects).
- E2E (tests/e2e/extension-loads.spec.ts) is a real Playwright MV3-load test that **SKIPS-with-reason** in this headless/sandbox environment (no display / unpacked-extension load unsupported); it executes the popup/options assertions only on a headful-capable host. Honest gap per §11.4.3, never a faked PASS.

- **Challenge:** challenges/extension/detect_and_forward_challenge.sh drives the real orchestrator + client end-to-end (detect → POST :7187/api/v1/download), PASS on captured evidence, mutation-verified (no-op stub → FAIL).
- **HelixQA bank:** challenges/helixqa-banks/boba-bobalink.yaml (symlinked from submodules/helixqa/banks/boba-bobalink.yaml) — http: cases require a LIVE merge-service on :7187; reports connection failure if down (no fail-open).

Legend

- **✓ path** — covered by the cited real test file (assertion on a user-observable outcome; fails against a no-op stub).
- **SKIP** — a real test exists but is operator-gated / topology-gated in this sandbox (e2e MV3 load; live-:7187 HelixQA).
- **PENDING** — no real test file yet (tracked in Gaps).
- **N/A** — test type does not apply to this client-side feature.

Autonomous-verification class (§11.4.52)

- **AUTONOMOUS_VERIFIED** — at least one autonomous path runs headlessly here with captured evidence + PASS/FAIL with no human in the loop.
- **AUTONOMOUS_DESIGNED** — an autonomous path exists/designed but its primary end-to-end proof is gated on an external dependency not present in this sandbox (live :7187, headful extension load).
- **OPERATOR_ATTENDED_ONLY** — the only proof path needs a host with a real display / operator (release blocker until promoted; tracked in Gaps).
- **NOT_APPLICABLE** — feature class does not warrant the path.

Coverage matrix

Feature / flow	unit	integration	security	stress
Magnet detection (BEP-9 parse, scan, highlight)	✓ tests/unit/magnet.test.ts, tests/unit/link-scanner.test.ts, tests/unit/text-	✓ tests/integration/pipeline.test.ts (STAGE 1)	PENDING	✓ test parser

Feature / flow	unit	integration	security	stress
	scanner.test.ts, tests/unit/ content.test.ts			
.torrent infohash (bencode → SHA-1)	✓ tests/unit/ torrent-file.test.ts, tests/unit/ bencode.test.ts	✓ tests/ integration/ pipeline.test.ts (STAGE 1 .torrent link)	PENDING	✓ test parser
Cross-scanner dedup (by lowercase infohash)	✓ tests/unit/ orchestrator.test.ts, tests/unit/scanner- base.test.ts	✓ tests/ integration/ pipeline.test.ts (STAGE 1: magnet×2 → 1)	PENDING	PENDING
Send → Boba :7187/api/v1/ download	✓ tests/unit/boba- client.test.ts, tests/unit/api- queue.test.ts	✓ tests/ integration/ pipeline.test.ts (STAGE 2/2b: captured request URL + body)	PENDING	PENDING
Offline FIFO queue (persist, retry, dead-letter, SW-restart)	✓ tests/unit/api- queue.test.ts	✓ tests/ integration/ pipeline.test.ts (STAGE 3/3b/3c persist + round- trip + drain)	PENDING	✓ test queue. (≥100 concu
Popup UI (detected list, Send, Send- All, status)	✓ tests/unit/ popup.test.ts, tests/ ally/ popup.ally.test.ts (9 ARIA)	—	PENDING	N/A
Options UI (7 tabs, save/ load config)	✓ tests/unit/ options.test.ts, tests/ally/ options.ally.test.ts (9 ARIA)	—	✓ tests/security/ secret- storage.test.ts	N/A
Background routing (SW message router,	✓ tests/unit/ background.test.ts, tests/unit/	—	PENDING	N/A

Feature / flow	unit	integration	security	stress
menus, badge, alarms)	background-token.test.ts			
Token decrypt-before-send (no embedded key, session passphrase)	✓ tests/unit/crypto.test.ts, tests/unit/boba-client-token.test.ts, tests/unit/background-token.test.ts	✓ tests/integration/pipeline.test.ts (STAGE 2c forward)	✓ tests/security/secret-storage.test.ts, tests/security/no-hardcoded-secret.test.ts	PENDING
Tab-group batch (dedup across group → one POST)	✓ tests/unit/tabgroups.test.ts, tests/unit/background.test.ts (MENU_SEND_GROUP)	—	✓ tests/security/manifest-least-privilege.test.ts (tabGroups minimal, no tabs)	PENDING
i18n (locale-key completeness)	✓ tests/unit/locale.test.ts	—	N/A	N/A
Manifest least-privilege / CSP (cross-cutting)	—	—	✓ tests/security/manifest-least-privilege.test.ts, tests/security/csp.test.ts	N/A

Gaps (highest-priority missing coverage — tracked follow-ups)

- 1. Live :7187 integration / end-to-end Challenge (Phase 4).** The send-flow integration substitutes the network boundary (fetchImpl stub) and the HelixQA bank (challenges/helixqa-banks/boba-bobalink.yaml) SKIPS when the merge-service is down. PENDING: a require_backend(7187) integration that actually POSTs to a running backend and asserts the torrent appears in qBittorrent (detect → send → server-side add), plus BE-1 (CORS) / BE-2 (.torrent multipart upload) backend work. Promotes “Send → :7187” from AUTONOMOUS_DESIGNED → AUTONOMOUS_VERIFIED.

2. **E2E real-extension load (Phase 8).** `tests/e2e/extension-loads.spec.ts` is real but `OPERATOR_ATTENDED_ONLY` in this sandbox (no MV3 unpacked-extension load; no display). PENDING: run on a headful-capable host so the popup/options/SW assertions actually execute, and add a real multi-tab tab-group e2e (Phase 5) and an offline-queue-durability e2e (SW termination).
3. **.torrent infohash end-to-end Challenge.** The detect→forward Challenge covers the magnet path; the .torrent-file forward path has no challenge- level captured-evidence run yet. PENDING.
4. **Additional locales + deeper-WCAG + theme evidence (Phase 6).** `locale.test.ts` guards only the en catalog; the plan targets 8 locales. Structural ARIA a11y IS now covered (`tests/a11y/popup.a11y.test.ts` + `options.a11y.test.ts` — 18 tests: roles / accessible-names / tablist↔tabpanel / live-regions, mutation-proven). PENDING: 7 more locale catalogs + a locale-switch render test, the deeper WCAG 2.1 AA checks a static HTML parse cannot cover (contrast ratios, focus order, keyboard traversal), and dark/light theme-switch evidence.
5. **Security depth (Phase 7).** Present: least-privilege manifest, CSP, no-hardcoded-secret, secret-storage. PENDING: §11.4.10.A pre-store credential-leak audit + pre-commit grep gate, a cred-leak-grep Challenge, and a fuller pen-test suite (magnet/dn XSS sanitization, `sender.id/sender.url` validation, log-redaction proof).
6. **Stress/chaos breadth beyond parsers + queue.** §11.4.85 `stress+chaos` exist for parsers and the offline queue. PENDING: `stress/chaos` for the scanner orchestrator (10k-link page, frame-budget), the BobaClient rate-limiter under API flood, and tab-group batching under many tabs.
7. **Per-store packaging + manual gate (Phase 9).** No `extension/ci-ext.sh` manual gate, per-store zips, bundle-size assertion, or §11.4.65 doc siblings yet. PENDING (NO CI/CD — manual only).